

Automated data collection at MAX IV: current status and future plans

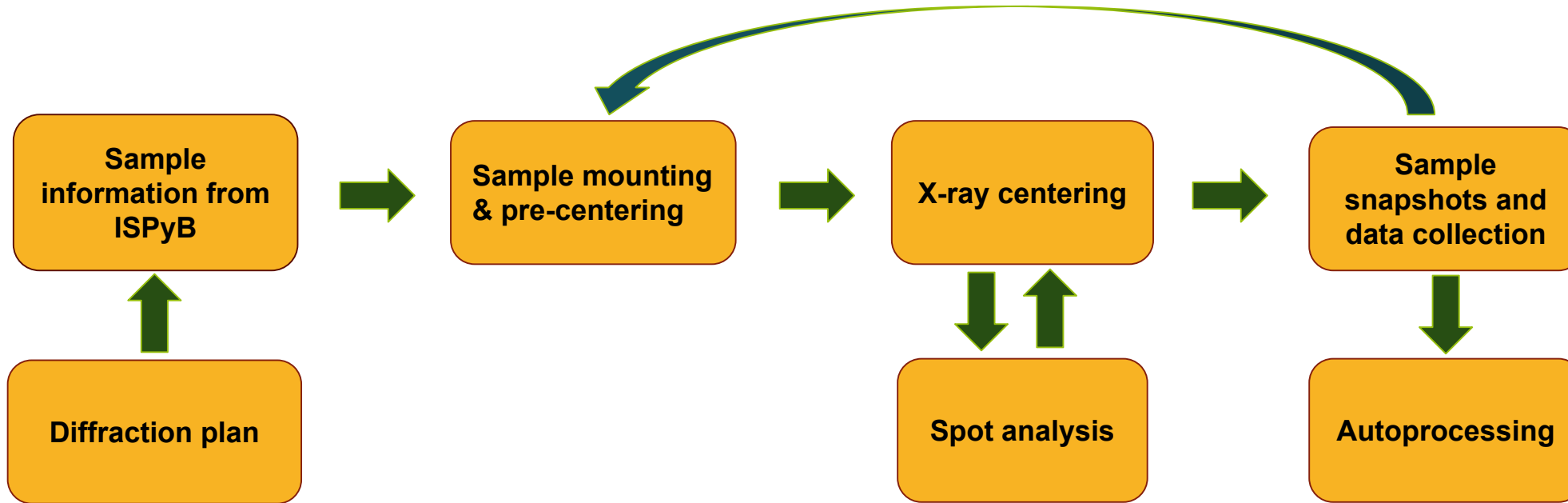


Joint MXCuBE/ISPyB Meeting June 2026
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Current status of automated data collection at MAX IV



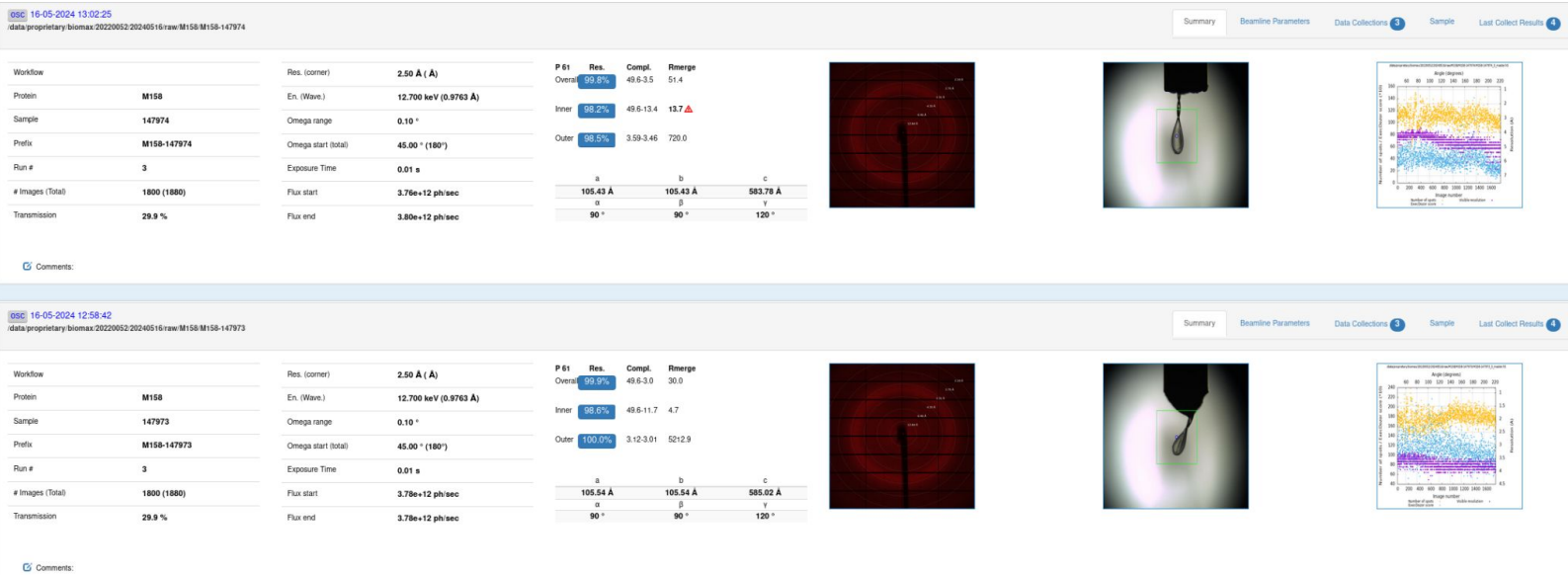
Current status of automated data collection at MAX IV



- Users get to choose the aimed resolution, energy, exposure time, rotation range (total and per image) for each sample, either in a diffraction plan (CSV file) uploaded to ISPyB or within MXCuBE.
- In-house loop autocentering program “loopfinder”. Sample drift detection.
- The parameters for X-ray centering are fixed: 30x30 grid, 10 micron cell and a 50 grid line scan at 90 degrees.
- Dozor is used for spot analysis, center is chosen based on the weighted dozor score over the beam size
- EdnaProc, fastdp and AutoProc launched after data collection.

Monitoring results

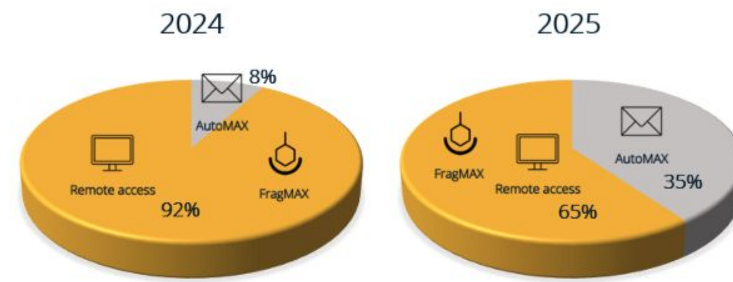
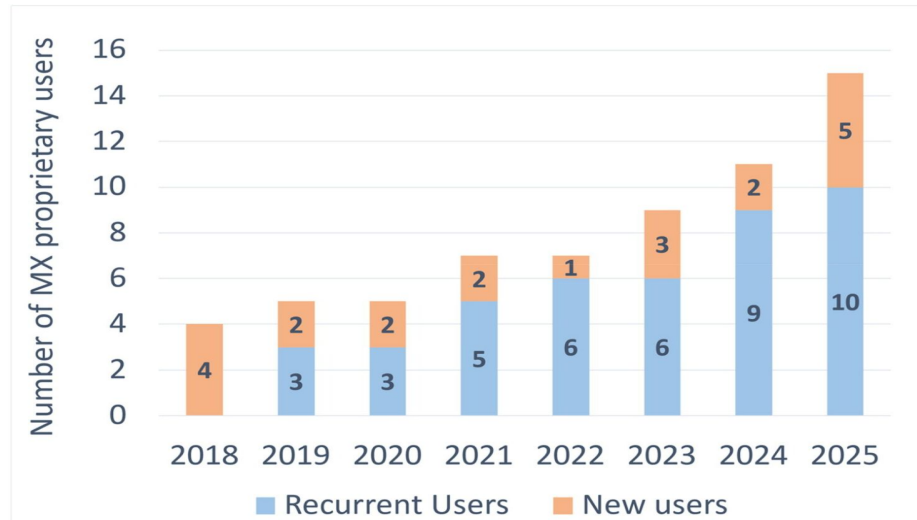
- A report is sent to the user every 20th sample.
- Results are shown in EXI.



- sample 3:05, time 20240516-102944, loaded True
- sample 3:06, time 20240516-103335, loaded True
 - sample 3:07, time 20240516-103727, loaded True
 - sample 3:08, time 20240516-104120, loaded True
 - sample 3:09, time 20240516-104511, loaded True
 - sample 3:10, time 20240516-104901, loaded True
 - sample 3:11, time 20240516-105252, loaded True
 - sample 3:12, time 20240516-105643, loaded True
 - sample 3:13, time 20240516-110034, loaded True
 - sample 3:14, time 20240516-110421, loaded True
 - sample 3:15, time 20240516-110811, loaded True
 - sample 3:16, time 20240516-111206, loaded True
 - sample 2:05, time 20240516-111557, loaded True
 - sample 2:06, time 20240516-112200, loaded True
 - sample 2:07, time 20240516-112842, loaded True
 - sample 2:08, time 20240516-113400, loaded True
 - sample 2:09, time 20240516-114039, loaded True
 - sample 5:01, time 20240516-115329, loaded True
 - sample 5:02, time 20240516-120934, loaded True
 - sample 5:03, time 20240516-121314, loaded True

Performance

- We can often run a dewarful of samples without the queue stopping (many errors can be handled by retrying or skipping to the next sample)
- Sample centering works surprisingly well. It outperforms click-to-center.
- Used by practically all FragMAX proposals, and a big proportion of proprietary proposals - usually within the mail-in access program “AutoMAX”. Slower but steady uptake among academic users.
- More efficient use of the beamtime: 61 shifts of automated data collection = 79 shifts of regularly scheduled beamtime



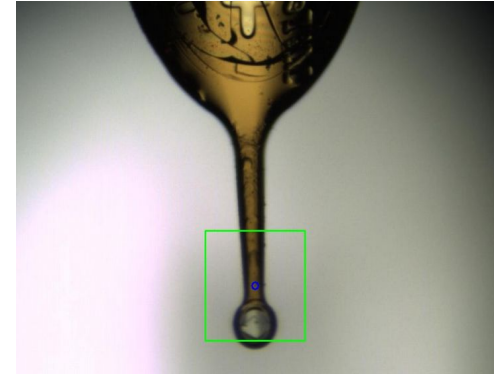
In 2026 the use is still up by 21% compared to June 2025

- AutoMAX is behind the large increase in industry usage at BioMAX since 2024

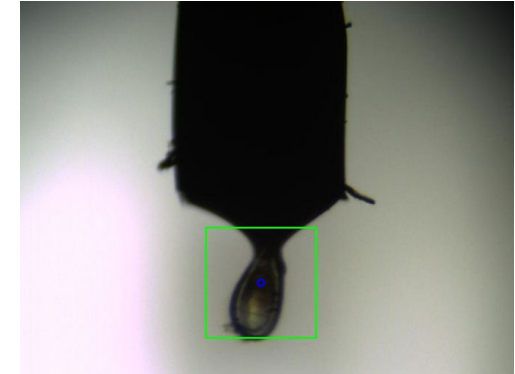
There is room for improvement!

“One size fits all” doesn’t...

- The approach does not work well for non uniform samples (in terms of sample size, shape or diffraction quality).
- The centering does not work well with multiple lattices. Because of the fixed scan area very large crystals do not fit the scan, scan the metal pin if the loop stem is too short...
- No sample information extracted from the centering scan.



OK



The pin may be hit during the scan

Operational issues

- Requires the MXCuBE interface for user log in and ISPyB communication.U1 slows down to a crawl when running long queues.
- X-ray centering and data collection are bundled as a single task in the queue.
- Implemented on a separate branch of MXCuBE3 (on end of life support)

OSC: 27-03-2025 01:36:42
/data/proprietary/BIOMAX/20250076/20250326/raw/PCSK9/PCSK9-MaxIV_2503_DRP-010_11

Prefix	Run	#images	Exposure Time	Res. (corner)	Wavelength	Transmission	Directory and Image template	Time	Run status
PCSK9-MaxIV_2503_DRP-010_11	1	30	0.01 s	2.0 Å (Å)	0.7293 Å	100%		01:36:49	Data collection successful
PCSK9-MaxIV_2503_DRP-010_11	2	50	0.1 s	2.0 Å (Å)	0.7293 Å	10.04%		01:37:52	Data collection successful
PCSK9-MaxIV_2503_DRP-010_11	3	3600	0.002 s	2.0 Å (Å)	0.7293 Å	100%		01:38:28	Data collection successful

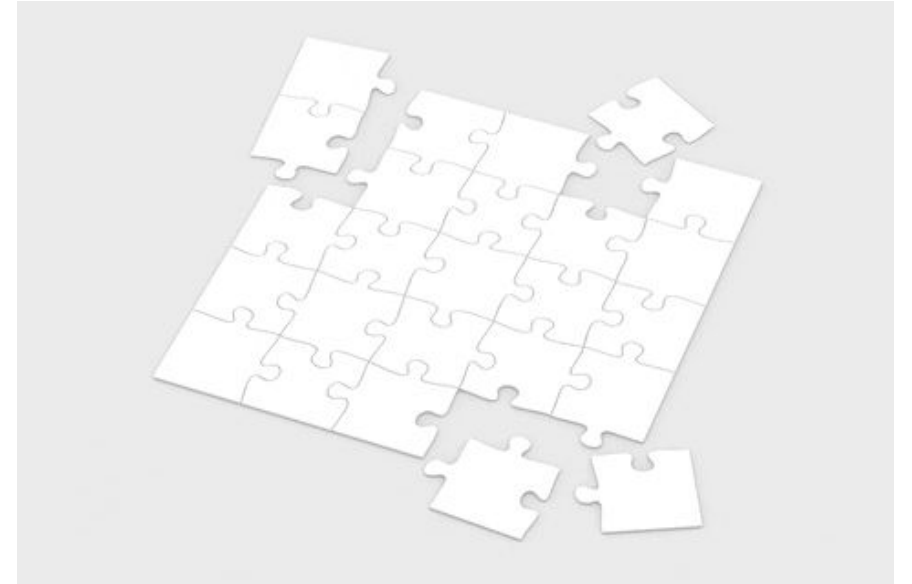
Future



Implementation in MXCuBE4

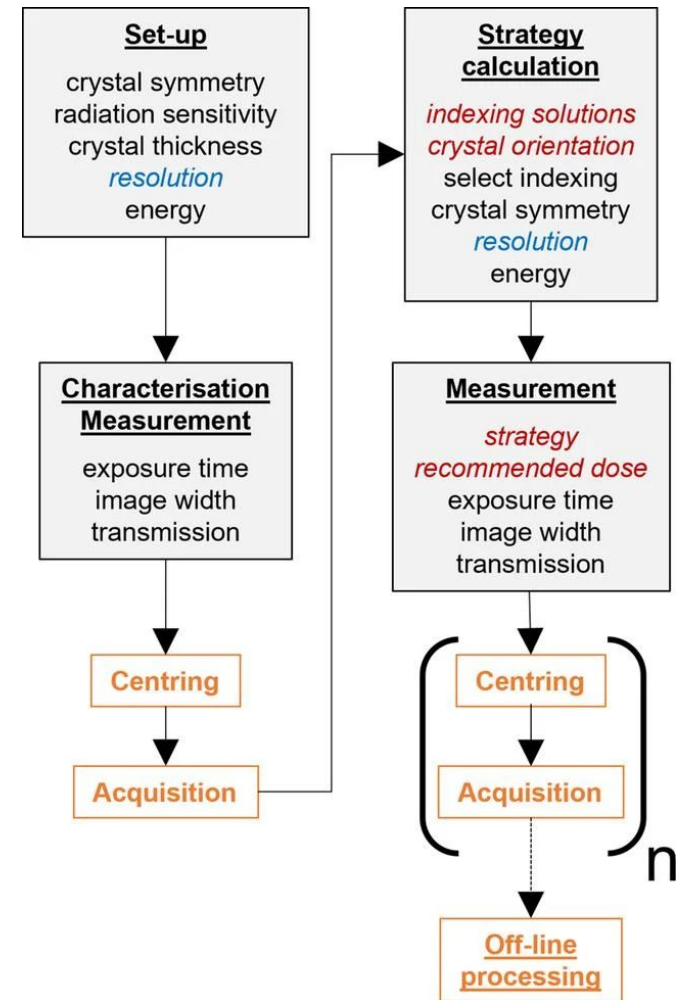
Plug and play

- Achieve a more modular approach
- Automated and “manual” data use the same code, no more swapping branches!
- Additional automated tasks can be offered to users, like XRF scans.
- Opens the door to reusing tasks in new workflows.



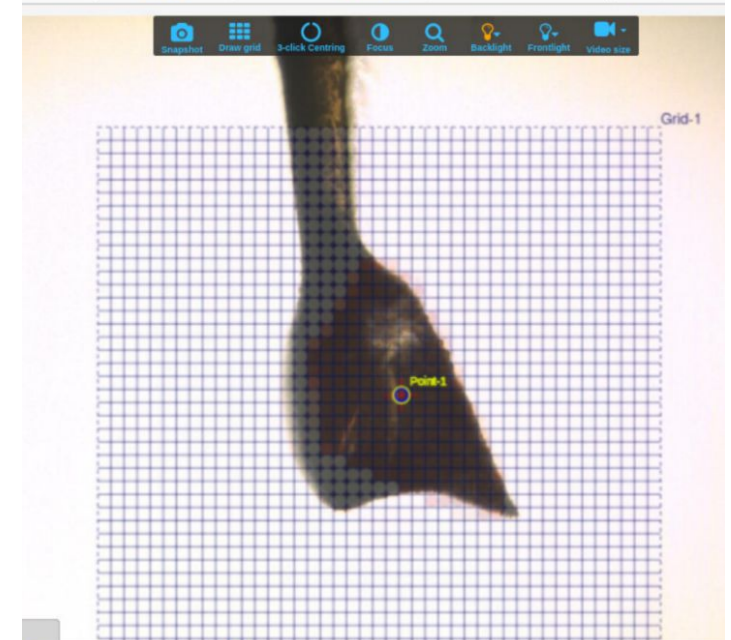
Expanding the workflow choice

- Implementation of automated GPhL workflows (using X-ray centering).
- So far we have the semiautomated workflows to work with click-to-center
- Strategy based on characterization
 - Dose budgeting
 - Equicomplete data
 - Phasing experiments
- Some work needed to present the results on py-ISPvB



Improve sample centering

- Participating in the automation WG project.
- Define bounding box based on optical centering results. We plan to evaluate Murko.
- Do two mesh scans at right-angles to get a better estimate of diffracting volume. This allows better matching of the beam size to crystal size
- Save heat maps and make the scan analysis results more accessible to users. Show them in py-ISPvB.



Beyond single crystal rotation

Automated fixed target SSX

- Custom ISARA sample grippers for mounting samples in capillaries and small grids at RT.
- Automated chip orientation (focus-based, could use fiducials too).
- Automated mesh scans over whole area of chip.
- Data processing using CrystFel pipeline developed at MicroMAX

Mesh and collect

- Suitable approach for multiple samples in loops, chips
- Challenge to recognize different crystals - indexing information or diffraction pattern comparison?



Acknowledgements

- Jie Nan, Mikel Eguiraun (implementation in MXCuBE3)
- Tobias Krojer (tireless tester)
- Isak Lindhe (loop finding)
- Piotr Goryl, Matheus Bernardi, Fabien Coronis, Elmir Jagudin, Bibiane Morakotkarn (MXCuBE4)
- Alberto Nardella (ISPyB integration)
- Aaron Finke (Edna2 integration)
- Gabriella Schröder, Josh Smith (multicrystal workflows)
- Cecilia Casadei (SSX processing pipelines)
- Monika Bjelcic, Swati Aggarwal, Ishkhan Gorgysian, Mirko Milas (“hot tools” for RT)

All present and past MX staff at MAX IV

Global Phasing L

MXCuBE collaboration

Thank you for your attention!

